Ahmedabad University

EXTENDING A MULTIMODAL PGM FOR BREAST CANCER SURVIVAL PREDICTION

CSE516 Probabilistic Graphical Models | Milestone 4 Final Presentation | Group G02

Dhanya Achari (AU2420163) Shifataj Shaikh (AU2420174) Paarth Jawaharani (AU2420056) Vishrut Patel (AU2340216)
Instructor: Prof. Dhaval Patel

THE PROBLEM

Rank breast cancer patients by survival risk using two views of the same tumour.

x Genomics

Gene expression: molecular state of the tumour

y Histology

Features from tissue slide images: tumour morphology

Output: a risk score per patient, scored with Harrell's C-index (50 = random, 100 = perfect ranking).

TCGA-BRCA cohort

974
patients

95
deaths observed (9.8%)

5
fixed train / val / test splits

585/146/243
patients per split

BASE PAPER: CURRENT STATE OF THE ART

Subramanian, Syeda-Mahmood and Do. Modelling-based joint embedding of histology and genomics using canonical correlation analysis for breast cancer survival prediction. *Artificial Intelligence in Medicine 149:102787, 2024.*

What the paper does

- Learns a joint embedding with penalised CCA: sparse CCA (SCCA) and graph-regularised SCCA (GN-SCCA)
- Extracts components with Hotelling (HD), projection (PD) or orthogonalised projection (OPD) deflation
- Feeds the embedding to a Cox model and reports C-index

What we found

- We reran the authors' code: 0.00 difference from their stored results
- CCA is used only as a projection: no p(z | x), no uncertainty, no principled missing modality

68.98

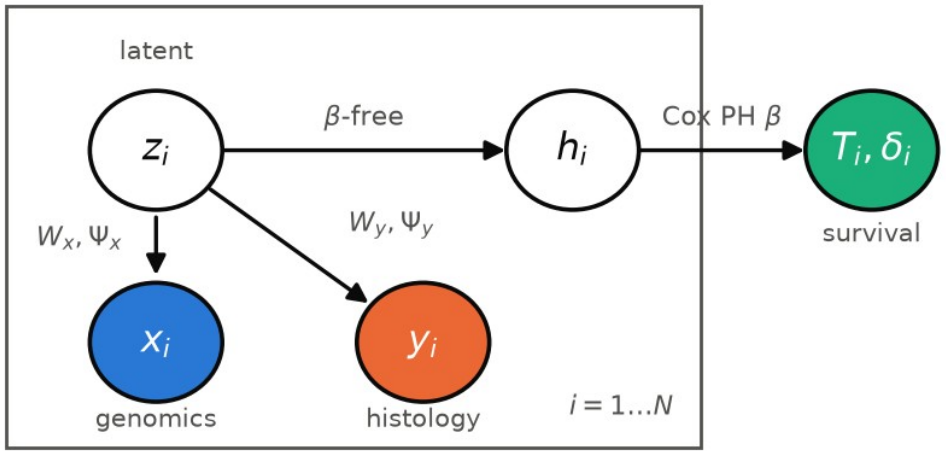
best C-index: SCCA/OPD, multimodal

MODEL, LEARNING AND INFERENCE

A generative latent-variable model in place of a projection

OUR MODEL: PROBABILISTIC CCA

Implemented graphical model (Group G02)
shaded = observed, unshaded = latent/derived; $X \perp\!\!\!\perp Y \mid Z$



Generative story

$$z \sim N(0, I_d)$$

$$x \mid z \sim N(W_x z + \mu_x, \Psi_x)$$

$$y \mid z \sim N(W_y z + \mu_y, \Psi_y)$$

- z: shared latent tumour state (d dimensions)
- Block-diagonal Ψ encodes X ⊥ Y | Z: all cross-modal dependence flows through z
- Survival enters downstream: Cox PH on the posterior mean of z

LEARNING: TWO ROUTES TO THE MLE

CCA-MLE

closed form (Bach and Jordan, 2005)

- Canonical directions U and correlations P from the sample covariances
- $W_x = \Sigma_{xx} U_x M_x$, $\Psi_x = \Sigma_{xx} - W_x W_x^T$ (same for y)
- Ridge 10^{-2} on Σ_{xx}, Σ_{YY}; one pass, no iterations

EM

our extension

- E-step: $E[z \mid x, y]$ and $E[zz^T \mid x, y]$ from the exact posterior
- M-step: $W = (\Sigma \tilde{o} E[z]^T)(\Sigma E[zz^T])^{-1}$, Ψ re-estimated and kept block-diagonal
- Random init (seed + fold), at most 300 iterations, tolerance 10^{-7}

Latent dimension d is chosen on the validation split only: CCA-MLE d = 30, EM d = 50 (grid 5, 10, 20, 30, 50).

INFERENCE: EXACT GAUSSIAN POSTERIORS

$p(z \mid x, y)$

both modalities

$$S = (I + W^T \Psi^{-1} W)^{-1}$$

$$m = S\, W^T \Psi^{-1} (o - \mu)$$

$p(z \mid x)$

genomics only

$$S_x = (I + W_x{}^T \Psi_x{}^{-1} W_x)^{-1}$$

$$m_x = S_x W_x{}^T \Psi_x{}^{-1} (x - \mu_x)$$

$p(z \mid y)$

histology only

$$S_y = (I + W_y{}^T \Psi_y{}^{-1} W_y)^{-1}$$

$$m_y = S_y W_y{}^T \Psi_y{}^{-1} (y - \mu_y)$$

Posterior mean m (= MAP) → **Cox PH, L2 penalty 0.1** → **Risk score $\beta^T m$** → **C-index on test split**

A missing modality just drops its block from W and Ψ: no imputation, and the posterior covariance widens to show the lost information.

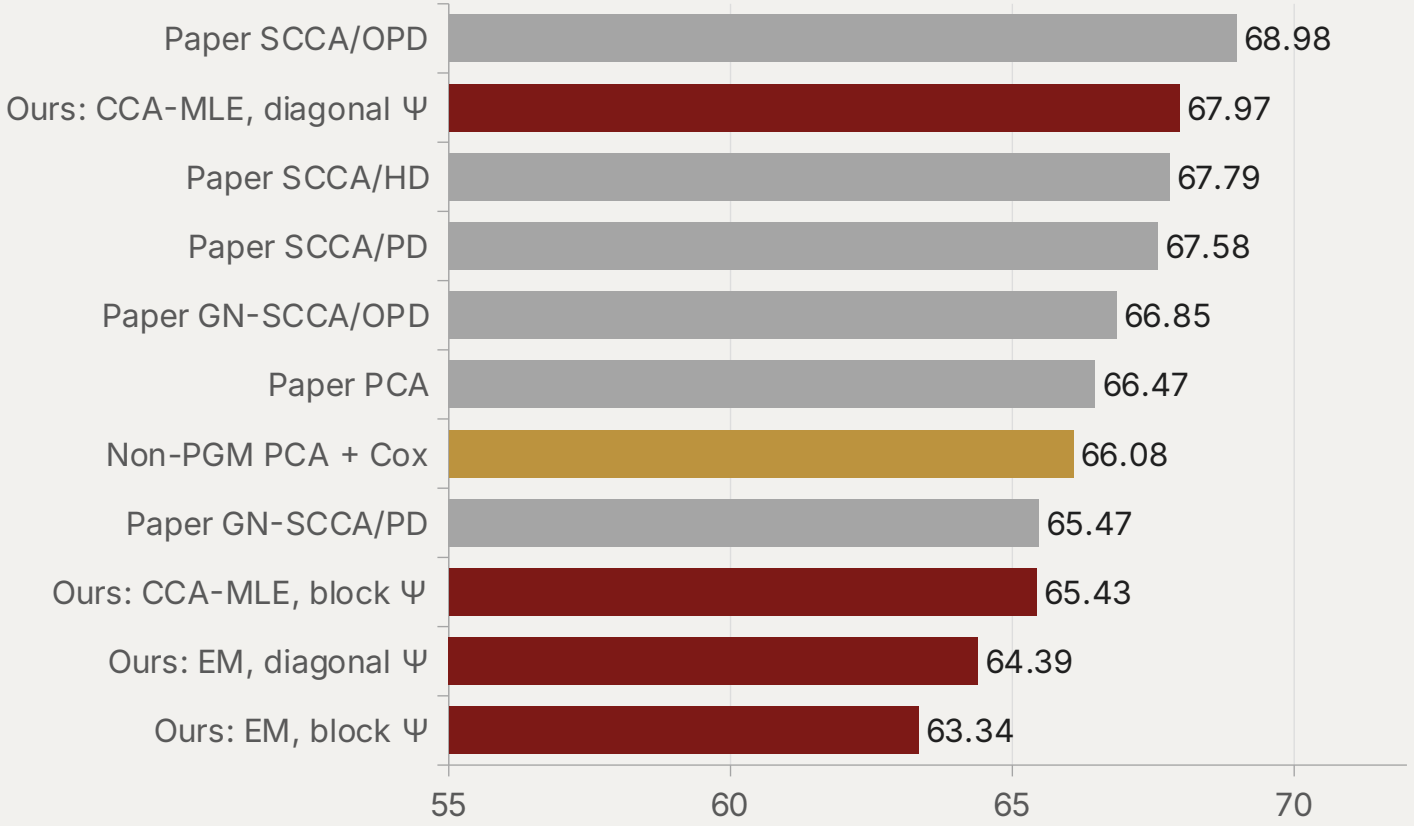
EXPERIMENTAL SETUP

1	**Data**	TCGA-BRCA, 974 patients, 95 deaths; the authors' 5 fixed splits (585 / 146 / 243)
2	**Input**	Paper's own 50 genomic + 50 imaging principal components, standardised on train
3	**Model selection**	d and all settings chosen on validation; test split used only for reporting
4	**Repeats**	5 seeds per split, averaged per split; mean ± SD (n − 1) over 5 splits
5	**Baselines**	Paper PCA, SCCA and GN-SCCA rerun with authors' code; non-PGM PCA with our Cox
6	**Statistics**	Paired t-test and Wilcoxon over splits, 95% CI of the paired difference

RESULTS

Discrimination, statistics, probability quality, robustness and cost

MULTIMODAL C-INDEX VS THE PAPER



Reading the chart

- Our best model (CCA-MLE, diagonal Ψ) is 1.01 points behind SCCA/OPD; paired p = 0.71
- Our block-Ψ models sit with the paper's PCA and GN-SCCA rows
- Mean of 5 splits; SD is about 7 points for every method (paper's SD convention gives 68.98 ± 6.16)

WHICH MODALITY CARRIES THE SIGNAL?

Method	Multimodal p(z|x,y)	Genomics p(z|x)	Histology p(z|y)
CCA-MLE, block Ψ	65.43 ± 7.36	67.31 ± 9.63	57.34 ± 7.22
EM, block Ψ	63.34 ± 7.53	69.13 ± 8.95	56.74 ± 9.50
CCA-MLE, diagonal Ψ	**67.97 ± 7.56**	**71.05 ± 9.62**	**56.93 ± 9.82**
EM, diagonal Ψ	64.39 ± 3.23	68.47 ± 9.63	62.56 ± 8.41
Non-PGM PCA + Cox	66.08 ± 7.79	67.82 ± 8.25	57.64 ± 9.49
Paper SCCA/OPD	68.98 ± 6.89	70.03 ± 8.38	55.86 ± 11.26

71.05

best genomics-only score of any method: CCA-MLE, diagonal Ψ

~57

histology alone reaches only about 57 for most methods

x alone

beats x + y in every row: adding histology lowers the C-index

ARE THE DIFFERENCES REAL?

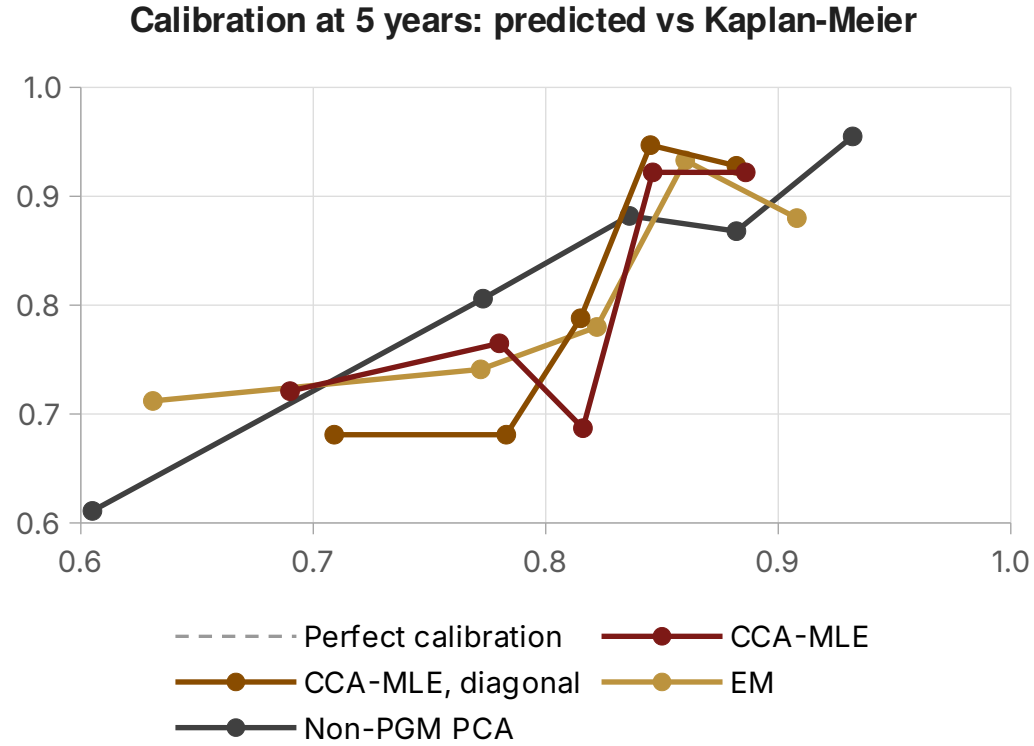
Paired comparison (multimodal)	Mean diff	95% CI	t-test p	Wilcoxon p	Wins
CCA-MLE − EM	+2.10	[−4.70, 8.90]	0.44	0.44	4 / 1
CCA-MLE − non-PGM PCA	−0.65	[−10.73, 9.43]	0.87	0.63	4 / 1
EM − non-PGM PCA	−2.75	[−7.53, 2.04]	0.19	0.31	1 / 4
CCA-MLE − paper SCCA/OPD	−3.55	[−13.36, 6.26]	0.37	0.44	2 / 3
CCA-MLE diagonal − paper SCCA/OPD	−1.01	[−8.07, 6.06]	0.71	1.00	3 / 2
CCA-MLE diagonal − CCA-MLE block	+2.54	[−1.86, 6.94]	0.18	0.31	4 / 1

Every 95% CI contains zero. With only 5 splits the smallest two-sided Wilcoxon p is 0.0625, so no comparison can pass $p < 0.05$ with that test. Our models are statistically tied with the paper.

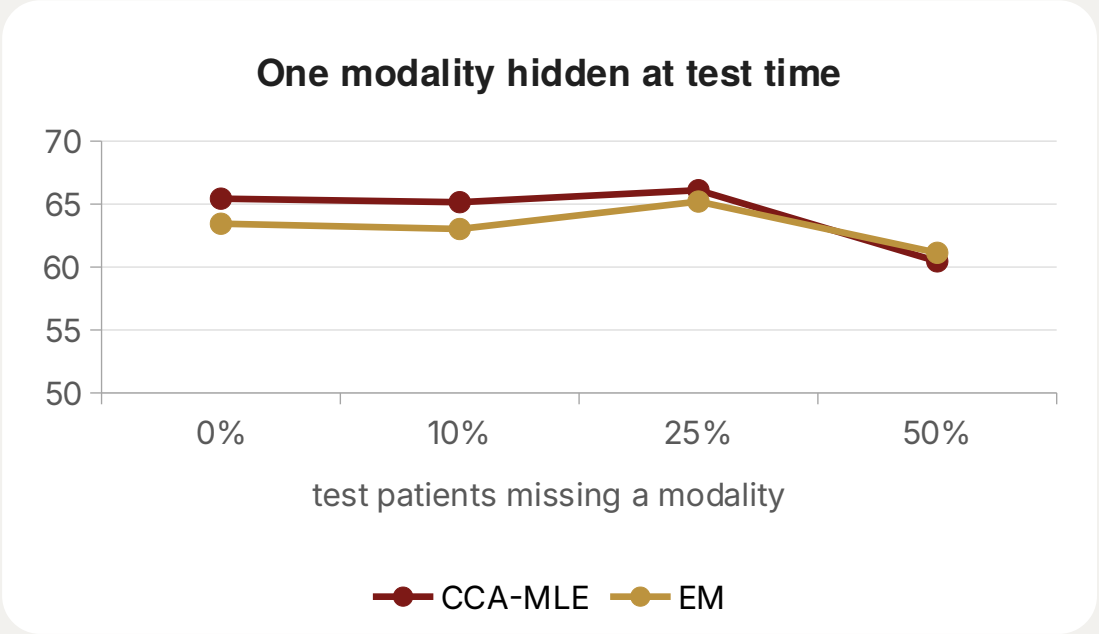
PROBABILITY QUALITY AND CALIBRATION

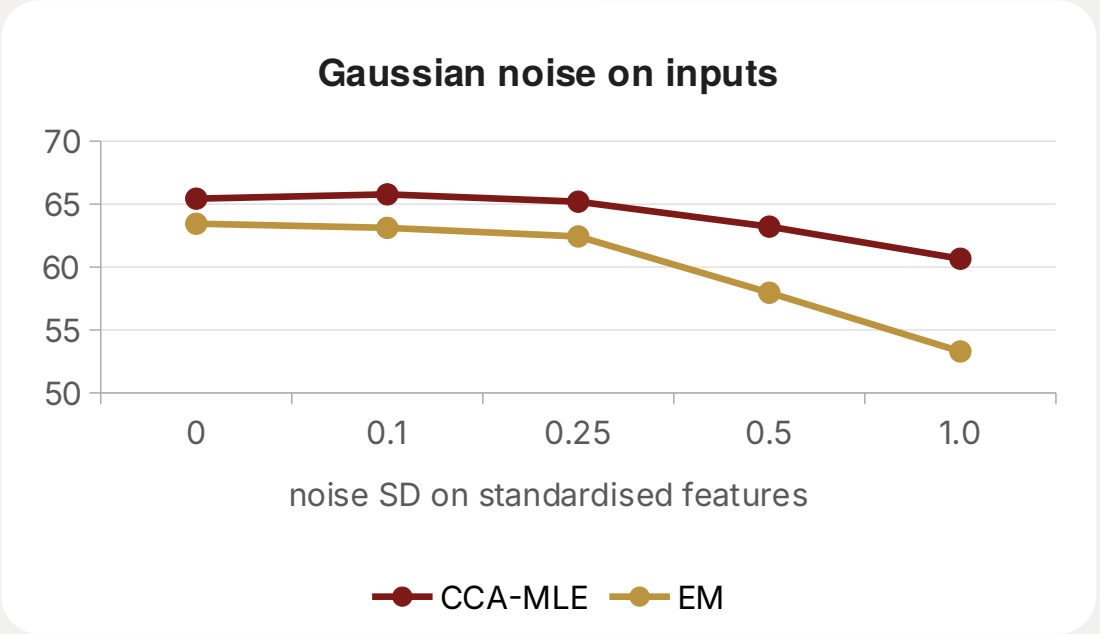
Method	IBS	Brier 3 y	Brier 5 y	ECE 5 y
CCA-MLE	0.0461	0.0814	0.1514	0.106
EM	0.0476	0.0856	0.1631	0.087
CCA-MLE, diagonal	0.0457	0.0803	0.1447	0.092
Non-PGM PCA	0.0469	0.0832	0.1528	0.078

- Lower is better for every column
- Diagonal CCA-MLE has the lowest Brier and IBS
- PCA is best calibrated; all gaps are small next to fold spread



ROBUSTNESS





- Up to 25% missing costs nothing: z is inferred from the remaining modality. CCA-MLE degrades more slowly under noise (60.7 vs 53.3 at SD 1).
- Training-size sweep (25 to 100%) is not monotonic: one subsample per fraction, so we treat it as noise.

ABLATIONS AND COMPUTATIONAL COST

Setting (multimodal C-index)	CCA-MLE	EM
Reference: block Ψ, ridge 10^{-2}, Cox 0.1	65.43	63.34
Diagonal Ψ	67.97	64.39
Ridge 10^{-1} / 10^{-3}	65.78 / 64.34	63.34 / 63.34
Cox penalty 0.01 / 1.0	65.50 / 64.35	62.21 / 62.32
EM initialised from CCA-MLE	–	62.31

Only the noise structure moves the result by more than a point; ridge and Cox penalty are flat.

Cost per fit (paper PCA input)

0.003 s

CCA-MLE, one pass

0.14 s

EM, about 92 iterations

157.1

CCA-MLE test NLL / patient

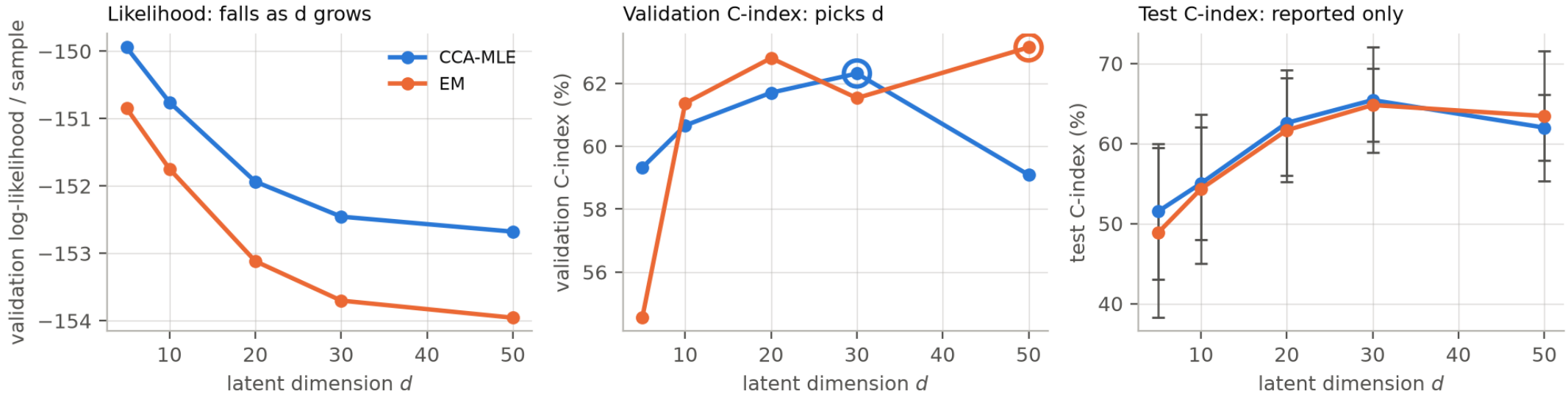
158.7

EM test NLL / patient

Diagonal-Ψ EM hits the 300-iteration cap on every split.

CHOOSING THE LATENT DIMENSION

Latent dimension sweep, input: paper's 50 + 50 PCA input. Circled = chosen on validation (cca-mle d = 30, em d = 50)



- Validation likelihood falls as d grows, so d is picked by validation C-index: d = 30 for CCA-MLE, d = 50 for EM. Test C-index was never used to choose.

WHERE WE STAND

68.98

Paper SCCA/OPD, best published configuration

67.97

Ours: CCA-MLE with diagonal Ψ

p = 0.71

paired difference −1.01, 95% CI [−8.07, 6.06]

What the PGM adds that the paper's projections cannot

1. Exact p(z | x) and p(z | y) with a covariance, so every risk score carries its uncertainty
2. Missing modality handled by marginalisation: no retraining, no imputation
3. A likelihood: models compared on held-out NLL, d chosen on principled grounds
4. Same accuracy class as the state of the art at a fraction of a second per fit

LIMITATIONS AND FUTURE WORK

Limitations

- 5 splits and 95 events give low statistical power
- Two-stage pipeline: survival is not a node in the graph
- Diagonal Ψ reuses d from the block-Ψ sweep; its EM hits the iteration cap
- Paper SCCA uses the full feature sets, ours the 50 + 50 PCA input
- Training-size sweep uses one subsample per fraction

Future work

- Put survival inside the PGM and learn z and β jointly
- Repeated cross-validation and an external cohort for power
- Tune d separately for diagonal Ψ
- Fit on the full genomic and histology feature sets
- Non-linear or sparse PCCA to close the remaining gap

REFERENCES AND REPRODUCIBILITY

[1] V. Subramanian, T. Syeda-Mahmood, M. N. Do. Modelling-based joint embedding of histology and genomics using CCA for breast cancer survival prediction. Artif. Intell. Med. 149:102787, 2024.

[2] F. R. Bach, M. I. Jordan. A probabilistic interpretation of canonical correlation analysis. Tech. Rep. 688, UC Berkeley, 2005.

[3] A. P. Dempster, N. M. Laird, D. B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. J. R. Stat. Soc. B 39(1):1–38, 1977.

[4] D. R. Cox. Regression models and life-tables. J. R. Stat. Soc. B 34(2):187–220, 1972.

[5] F. E. Harrell et al. Evaluating the yield of medical tests. JAMA 247(18):2543–2546, 1982.

[6] E. Graf et al. Assessment and comparison of prognostic classification schemes for survival data. Stat. Med. 18:2529–2545, 1999.

Reproduce everything

github.com/shifaa174/CSE-516--PGM_Healthcare_brca

From the repository root:

```
python M4_G02/Code/run_all.py .
```

Regenerates baselines, 5-seed results, ablations, robustness, statistics, probability quality and every figure in M4_G02/Results.



Ahmedabad University

THANK YOU

Group G02 | CSE516 Probabilistic Graphical Models | Ahmedabad University